TASK-11:

```
varshini@Varshini:~$ mkdir task11_pdf
varshini@Varshini:~$ cd task11_pdf
varshini@Varshini:~/task11_pdf$ nano pdf_open.c
varshini@Varshini:~/task11_pdf$ nano pdf_open.c
varshini@Varshini:~/task11_pdf$ ls
pdf_open.c
varshini@Varshini:~/task11_pdf$ cd ~/Downloads
-bash: cd: /home/varshini/Downloads: No such file or directory
varshini@Varshini:~/task11_pdf$ ls
pdf_open.c
varshini@Varshini:~/task11_pdf$ cp pdf_open.pdf sample.pdf
cp: cannot stat 'pdf_open.pdf': No such file or directory
varshini@Varshini:~/task11_pdf$ cp pdf_open.c sample.pdf
varshini@Varshini:~/task11_pdf$ ls -l sample.pdf
-rw-r--r-- 1 varshini varshini 970 Aug 13 04:00 sample.pdf
varshini@Varshini:~/task11_pdf$ ls
pdf_open.c  sample.pdf
varshini@Varshini:~/task11_pdf$ gcc pdf_open.c -o pdf_open
varshini@Varshini:~/task11_pdf$ ./pdf_open
Browser process started.
Browser PID: 1396

Parent/browser process continues running.
Parent PID: 1396
Child/PDF viewer PID: 1397

Child process created.
Child PID: 1397
Opening PDF using xdg-open...
exec failed: No such file or directory

PDF viewer process finished.
Browser process continues.
varshini@Varshini:~/task11_pdf$ ps -ef | grep pdf_open
varshini    1410    1196  0 04:03 pts/0    00:00:00 grep --color=auto pdf_open
varshini@Varshini:~/task11_pdf$
```

TASK-12:

```
varshini@Varshini:~$ mkdir task12
varshini@Varshini:~$ cd task12
varshini@Varshini:~/task12$ nano shell_loop.c
varshini@Varshini:~/task12$ ^C
varshini@Varshini:~/task12$ gcc shell_loop.c -o shell_loop
varshini@Varshini:~/task12$ ./shell_loop
Simple Linux Command Loop
Type 'exit' to quit.

myshell>
myshell> hello
You entered: hello
myshell> linux
You entered: linux
myshell> existing shell
You entered: existing shell
myshell>
```

TASK-13:

```
varshini@Varshini:~$ cd ~
varshini@Varshini:~$ pwd
/home/varshini
varshini@Varshini:~$ mkdir task13
mkdir: task13: File exists
varshini@Varshini:~$ cd task13
varshini@Varshini:~/task13$ pwd
/home/varshini/task13
varshini@Varshini:~/task13$ nano task13.c
varshini@Varshini:~/task13$ ls
shell_loop  shell_loop.c  task13.c
varshini@Varshini:~/task13$ gcc task13.c -o task13
varshini@Varshini:~/task13$ ls
shell_loop  shell_loop.c  task13  task13.c
varshini@Varshini:~/task13$ ./task13
myshell> hello
Hello! Welcome to MyShell.
myshell> help

Available commands:
  hello - Display greeting
  help  - Display available commands
  exit  - Exit the shell
myshell> abc
Unknown command: abc
myshell> exit
Exiting shell...
varshini@Varshini:~/task13$
```

TASK-14:

```
varshini@Varshini:~$ mkdir task14
varshini@Varshini:~$ cd task14
varshini@Varshini:~/task14$ pwd
/home/varshini/task14
varshini@Varshini:~/task14$ nano task14.c
varshini@Varshini:~/task14$ gcc task14.c -o task14
varshini@Varshini:~/task14$ ./task14
myshell> heloo
You entered: heloo
myshell> world
You entered: world
```

```
myshell> history
1  heloo
2  world
myshell> hello960
You entered: hello960
myshell> hello
You entered: hello
myshell> exit
Exiting shell...
varshini@Varshini:~/task14$
```

TASK-15:

```
varshini@Varshini:~/task14$ cd ..
varshini@Varshini:~$ mkdir task15
varshini@Varshini:~$ cd task15
varshini@Varshini:~/task15$ nano task15.c
varshini@Varshini:~/task15$ gcc task15.c -o task15
varshini@Varshini:~/task15$ touch main.c test.c program.txt
hello.c
varshini@Varshini:~/task15$ ls
hello.c  main.c  program.txt  task15  task15.c  test.c
varshini@Varshini:~/task15$ ./task15
-rw-r--r-- 1 varshini varshini     0 Sep  9 05:24 hello.c
-rw-r--r-- 1 varshini varshini     0 Sep  9 05:24 main.c
-rw-r--r-- 1 varshini varshini  1379 Sep  9 05:23 task15.c
-rw-r--r-- 1 varshini varshini     0 Sep  9 05:24 test.c
Parent: Both processes completed.
varshini@Varshini:~/task15$
```

TASK-16:

```
varshini@Varshini:~$ cd ~
mkdir -p task16
cd task16
varshini@Varshini:~/task16$
varshini@Varshini:~/task16$ cat > server.c <<'EOF'
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <sys/wait.h>
#include <signal.h>
#include <string.h>

#define MAX_CLIENTS 3
#define BUFFER_SIZE 256

char c2s[MAX_CLIENTS][50] = {
    "client1_to_server",
    "client2_to_server",
    "client3_to_server"
};

char s2c[MAX_CLIENTS][50] = {
    "server_to_client1",
    "server_to_client2",
    "server_to_client3"
};

pid_t children[MAX_CLIENTS];

void cleanup(int sig)
{
    int i;

    printf("\nServer shutting down...\n");

    for (i = 0; i < MAX_CLIENTS; i++)
    {
        if (children[i] > 0)
```

```
    }

    for (i = 0; i < MAX_CLIENTS; i++)
    {
        unlink(c2s[i]);
        unlink(s2c[i]);
    }

    exit(0);
}

void handle_usr1(int sig)
{
    printf("\nServer received SIGUSR1.\n");
    fflush(stdout);
}

void handle_chld(int sig)
{
    while (waitpid(-1, NULL, WNOHANG) > 0)
    {
    }
}

void client_handler(int client_id)
{
    int read_fd;
    int write_fd;
    char buffer[BUFFER_SIZE];
    char response[BUFFER_SIZE];

    read_fd = open(c2s[client_id], O_RDONLY);

    if (read_fd < 0)
    {
        perror("open client-to-server FIFO");
        exit(1);
    }

    write_fd = open(s2c[client_id], O_WRONLY);
```

```
    if (write_fd < 0)
    {
        perror("open server-to-client FIFO");
        close(read_fd);
        exit(1);
    }

    printf("Handler %d started. PID = %d\n",
           client_id + 1, getpid());
    fflush(stdout);

    while (1)
    {
        memset(buffer, 0, sizeof(buffer));

        ssize_t n = read(read_fd, buffer, sizeof(buffer) - 1);

        if (n <= 0)
            break;

        buffer[n] = '\0';

        if (strcmp(buffer, "exit") == 0)
        {
            snprintf(response, sizeof(response),
                     "Server: Client %d disconnected.",
                     client_id + 1);

            write(write_fd, response, strlen(response) + 1);
            break;
        }

        printf("Client %d: %s\n", client_id + 1, buffer);
        fflush(stdout);

        snprintf(response, sizeof(response),
                 "Server received from Client %d: %s",
                 client_id + 1, buffer);

        write(write_fd, response, strlen(response) + 1);
    }
```

```
        write(write_fd, response, strlen(response) + 1);
    }

    close(read_fd);
    close(write_fd);

    printf("Handler %d exiting.\n", client_id + 1);
    fflush(stdout);

    exit(0);
}

int main()
{
    int i;

    signal(SIGUSR1, handle_usr1);
    signal(SIGINT, cleanup);
    signal(SIGCHLD, handle_chld);

    printf("===== Multi-Client FIFO Server =====\n");

    /* Create all FIFOs */
    for (i = 0; i < MAX_CLIENTS; i++)
    {
        mkfifo(c2s[i], 0666);
        mkfifo(s2c[i], 0666);
    }

    printf("All FIFOs created.\n");

    /* Create one handler for each client */
    for (i = 0; i < MAX_CLIENTS; i++)
    {
        children[i] = fork();

        if (children[i] < 0)
        {
            perror("fork");
            cleanup(SIGINT);
        }
```

```
    while (1)
    {
        pause();
    }

    return 0;
}
EOF
varshini@Varshini:~/task16$
varshini@Varshini:~/task16$ cat > client.c <<'EOF'
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <signal.h>
#include <string.h>

#define BUFFER_SIZE 256

char c2s[3][50] = {
    "client1_to_server",
    "client2_to_server",
    "client3_to_server"
};

char s2c[3][50] = {
    "server_to_client1",
    "server_to_client2",
    "server_to_client3"
};

int write_fd;
int read_fd;

void handle_sigusr1(int sig)
{
    printf("\nClient received SIGUSR1.\n");
    fflush(stdout);
}

void handle_sigint(int sig)
```

```c
void handle_sigint(int sig)
{
    printf("\nClient exiting...\n");

    close(write_fd);
    close(read_fd);

    exit(0);
}

int main(int argc, char *argv[])
{
    int client_id;
    char buffer[BUFFER_SIZE];
    char response[BUFFER_SIZE];

    if (argc != 2)
    {
        printf("Usage: %s <client_number>\n", argv[0]);
        printf("Example: %s 1\n", argv[0]);
        return 1;
    }

    client_id = atoi(argv[1]) - 1;

    if (client_id < 0 || client_id > 2)
    {
        printf("Client number must be 1, 2, or 3.\n");
        return 1;
    }

    signal(SIGUSR1, handle_sigusr1);
    signal(SIGINT, handle_sigint);

    printf("Client %d started. PID = %d\n",
           client_id + 1, getpid());

    /*
     * Open client-to-server FIFO for writing.
     * The server handler opens the read end.
```

```
     */
    write_fd = open(c2s[client_id], O_WRONLY);

    if (write_fd < 0)
    {
        perror("open client-to-server FIFO");
        return 1;
    }

    /*
     * Open server-to-client FIFO for reading.
     */
    read_fd = open(s2c[client_id], O_RDONLY);

    if (read_fd < 0)
    {
        perror("open server-to-client FIFO");
        close(write_fd);
        return 1;
    }

    printf("Connected to server.\n");

    while (1)
    {
        printf("Client %d > ", client_id + 1);
        fflush(stdout);

        if (fgets(buffer, sizeof(buffer), stdin) == NULL)
            break;

        buffer[strcspn(buffer, "\n")] = '\0';

        if (strlen(buffer) == 0)
            continue;

        write(write_fd, buffer, strlen(buffer) + 1);

        memset(response, 0, sizeof(response));

        if (read(read_fd, response, sizeof(response) - 1) <= 0)
```

```
        if (read(read_fd, response, sizeof(response) - 1) <= 0)
            break;

        printf("%s\n", response);

        if (strcmp(buffer, "exit") == 0)
            break;
    }

    close(write_fd);
    close(read_fd);

    printf("Client %d terminated.\n", client_id + 1);

    return 0;
}
EOF
varshini@Varshini:~/task16$ gcc -Wall -Wextra server.c -o server
server.c: In function ‘cleanup’:
server.c:28:18: warning: unused parameter ‘sig’ [-Wunused-parameter]
   28 | void cleanup(int sig)
      |              ~~~~^~~
server.c: In function ‘handle_usr1’:
server.c:49:22: warning: unused parameter ‘sig’ [-Wunused-parameter]
   49 | void handle_usr1(int sig)
      |                  ~~~~^~~
server.c: In function ‘handle_chld’:
server.c:55:22: warning: unused parameter ‘sig’ [-Wunused-parameter]
   55 | void handle_chld(int sig)
      |                  ~~~~^~~
server.c: In function ‘client_handler’:
server.c:115:51: warning: ‘%s’ directive output may be truncated writing up to 255 bytes into a region of size between 215 and 225 [-Wformat-truncation]
  115 |                 "Server received from Client %d: %s",
      |                                                   ^~
  116 |                 client_id + 1, buffer);
      |                                ~~~~~~
server.c:114:9: note: ‘snprintf’ output between 32 and 297 bytes into a destination of size 256
  114 |         snprintf(response, sizeof(response),
      |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
  115 |                 "Server received from Client %d: %s",
```

```
      |                                   ~~~~~~~~~~~~~~~~~~~~~~
varshini@Varshini:~/task16$ ls -l server
-rwxr-xr-x 1 varshini varshini 17384 Sep  9 05:54 server
varshini@Varshini:~/task16$ ./server
===== Multi-Client FIFO Server =====
All FIFOs created.
Server PID: 897
Waiting for clients...
Handler 1 started. PID = 898
Client 1: hello
Client 1: cd -/task16
Handler 2 started. PID = 899
Client 2: hello from client 2
Handler 3 started. PID = 900
Client 3: hello from client 3
```

```
varshini@Varshini:~$ cd ~/task16
gcc -Wall -Wextra client.c -o client
client.c: In function ‘handle_sigusr1’:
client.c:25:25: warning: unused parameter ‘sig’ [-Wunused-parameter]
   25 | void handle_sigusr1(int sig)
      |                     ~~~~^~~
client.c: In function ‘handle_sigint’:
client.c:31:24: warning: unused parameter ‘sig’ [-Wunused-parameter]
   31 | void handle_sigint(int sig)
      |                    ~~~~^~~
varshini@Varshini:~/task16$ ./client 1
Client 1 started. PID = 965
Connected to server.
Client 1 > hello
Server received from Client 1: hello
Client 1 > cd -/task16
Server received from Client 1: cd -/task16
Client 1 >
```

```
varshini@Varshini:~$ cd ~/task16
./client 2
Client 2 started. PID = 1008
Connected to server.
Client 2 > hello from client 2
Server received from Client 2: hello from client 2
Client 2 >
```

```
varshini@Varshini:~$ cd ~/task16
./client 3
Client 3 started. PID = 1052
Connected to server.
Client 3 > hello from client 3
Server received from Client 3: hello from client 3
Client 3 >
```

```
varshini@Varshini:~$ cd ~/task16
ls -l
total 48
-rwxr-xr-x 1 varshini varshini 17296 Sep  9 05:55 client
-rw-r--r-- 1 varshini varshini  2376 Sep  9 05:54 client.c
prw-r--r-- 1 varshini varshini     0 Sep  9 05:58 client1_to_server
prw-r--r-- 1 varshini varshini     0 Sep  9 05:58 client2_to_server
prw-r--r-- 1 varshini varshini     0 Sep  9 05:58 client3_to_server
-rwxr-xr-x 1 varshini varshini 17384 Sep  9 05:54 server
-rw-r--r-- 1 varshini varshini  3238 Sep  9 05:54 server.c
prw-r--r-- 1 varshini varshini     0 Sep  9 05:58 server_to_client1
prw-r--r-- 1 varshini varshini     0 Sep  9 05:58 server_to_client2
prw-r--r-- 1 varshini varshini     0 Sep  9 05:58 server_to_client3
varshini@Varshini:~/task16$
```